Rice's Formula for Event-Driven Networks: Predicting and Training Temporal Sparsity via Differentiable Level-Crossing Rates

Gunner Levi Howe
Independent Researcher
gunnerlevihowe@gmail.com

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Abstract

Delta networks spend compute only when an activation changes by more than a threshold: each supra-threshold change is an *event* that triggers the corresponding weight-column MACs and memory traffic. A delta event is, definitionally, a level crossing of the activation time series—yet the delta-network literature tunes thresholds by grid search and only ever *measures* event counts after the fact. We connect this literature to the century-old analytic theory of level crossings. **Predict:** on three trained networks (GRU keyword spotting on Speech Commands, a row-sequential-MNIST GRU, an enwik8 char-transformer), an empirical crossing-rate profile fitted on a calibration split predicts per-channel crossing rates on held-out data to 3.1%–2.9% median relative error, and open-loop send-on-delta simulation on cached traces predicts closed-loop delta-network event rates to within 3.0% (SC2) in the regime where the delta approximation preserves accuracy. Rice's Gaussian formula works where activations are near-Gaussian (LayerNorm-fed streams) and fails predictably with kurtosis (GELU outputs, deep GRU states); an iid baseline fails by 5.6×–17.3×. **Allocate:** inverting the predicted rate curves turns a global event budget into per-stream thresholds analytically; on Speech Commands this matches the Pareto front of a 48-configuration random search at ∼2% of its tuning cost, and realized event rates land within 4% of target. **Train (negative):** fine-tuning against a one-sided differentiable crossing budget makes traces comply (crossings drop as demanded, increment scales shrink 3.9×) but does *not* improve the accuracy–events Pareto front over post-hoc thresholding of the unmodified network—and neither do L1-on-deltas or activity-rate controls. The mechanism is an approximate scale-freeness: send-on-delta event rates depend on θ/σ_δ, so smoothness training rescales dynamics without restructuring them, while fixed-threshold comparisons misleadingly report large savings (36% fewer events at matched absolute θ). Prediction and allocation work; training-for-temporal-sparsity, in these tasks, does not beat the post-hoc baseline it is meant to replace. We release estimators, receipts, and benchmarks.

1 Introduction

Delta networks—delta-RNNs, sigma–delta networks, event-driven transformers—spend compute only when an activation *changes* by more than a threshold θ: each supra-threshold change transmits a delta, triggers the corresponding weight-column MACs and memory traffic, and is therefore a quantum of energy [2, 5, 17, 18, 26]. In this literature the thresholds are tuned by post-hoc accuracy-vs-speedup sweeps and the event counts are always *measured after the fact*; no published delta-network method predicts event rates from activation statistics or trains against an analytic rate objective.

A delta event is, definitionally, a level crossing of the activation time series. Level-crossing statistics have an analytic theory that is eighty years old: Rice's formula [10, 21] gives the mean crossing rate of a stationary Gaussian process at any level from two spectral moments, and its smoothed Monte-Carlo estimator—built and validated as a differentiable training objective in the preceding papers of this program [7–9]—needs no Gaussianity at all. This paper connects the two. It is the fourth paper of the program, and the arc matters: Howe [7] validated a one-sided crossing *budget* whose loss and gradient vanish within budget, then reported honestly that PINNs fail by under- rather than over-oscillating, so the budget was inert there. Delta networks are the budget's native domain—excess crossings are excess energy *by construction*—and this paper is the test of that promise.

The results split cleanly into two positives and one negative, and we report all three:

1. **Validated estimator (E0).** The temporal segment estimator (exact for piecewise-linear traces, differentiable, converging to hard counts) and Rice predictions agree with ground truth to $\leq$1.9% on synthetic processes with known spectra; the exact discrete-time Gaussian crossing formula (an Owen's-T expression) removes the up-to-8.8% bias of the continuous formula on coarsely sampled sequences. The send-on-delta event rate is bounded by total variation over θ; the gap—hysteresis of the anchor memory—is a strikingly process-independent function $\gamma(\theta/\sigma_\delta)$ for $\theta \lesssim \sigma_\delta/2$ (spread 0.5% across four process families).

2. **Prediction and allocation on real networks (E1–E2).** A calibration-split crossing profile predicts held-out per-channel crossing rates to 3.1%–2.9% median error across a GRU keyword spotter, a row-sequential-MNIST GRU, and a char-transformer; Rice's Gaussian form holds on near-Gaussian streams and fails predictably with kurtosis. Open-loop simulation on cached traces predicts deployed (closed-loop) event rates to a few percent in the regime where accuracy survives. Inverting these curves allocates per-stream thresholds analytically: on Speech Commands the resulting operating points match a 48-configuration random search at $\sim$2% of its tuning cost.

3. **An honest negative with a mechanism (E3–E4).** Fine-tuning against the one-sided crossing budget makes networks comply—crossing profiles drop as demanded, increment scales shrink 3.9×—but the accuracy–events Pareto front does *not* improve over post-hoc thresholding of the unmodified network. Neither do the obvious alternatives (L1 on activation deltas, activity-rate regularization, plain fine-tuning). The mechanism: send-on-delta event rates are approximately scale-free—they depend on the threshold in units of the increment scale, θ/σ_δ—so smoothness training *rescales* dynamics without *restructuring* them. At matched absolute threshold the budget-trained network fires 36% fewer events, which is exactly the comparison a fixed-threshold evaluation would report as a win; at matched *events* the base network is as good or better. Post-hoc thresholding is a much stronger baseline than the training literature assumes.

2 Related work

Delta and sigma–delta networks. Neil et al. [17] threshold changes of RNN inputs and hidden states against an anchor memory of the last transmitted value and accumulate matrix–vector products incrementally; the mechanism transfers to embedded video [26], to silicon [2, 5], and to recent RF applications [24], and underlies sigma–delta quantized networks [18]. Thresholds are set by accuracy-vs-speedup sweeps throughout, and event counts are reported only after measurement. A parallel *training*-side line adds delta/activation-change penalties during training and reports temporal-sparsity gains—the Delta Activation Layer [25, 26], and EGRU [22], which trains activity-sparse GRUs with surrogate gradients through a hard threshold. Our E3–E4 results speak directly to this line (§8): at matched, re-tuned event budgets we do not reproduce a training advantage, and §8 explains why the fixed-threshold evaluations these methods use overstate it.

Spiking transformers and spatial activation sparsity. Spikformer [27] and SpikeGPT [28] use spike *coding*, a different mechanism from delta thresholding; we cite and do not claim their ground. The activation-sparsity line [14] concerns *spatial* sparsity (which units are zero); ours is *temporal* sparsity (which units change).

Level-crossing theory in ML and signal processing. Rice's formula appears in machine learning in loss-landscape critical-point counting [1, 4] and in our own program [7–9]—never as an activation-event or energy model. Zero-crossing statistics are classical discrimination features [11, 12]; send-on-delta sampling theory [16] bounds event rates by total variation. The novelty here, as in the companion papers, is the *instantiation*: crossing rates as a differentiable model of, and budget on, delta-network events.

3 Theory: delta events as level crossings

Setting. Let $a_c(t)$, $t = 1, \dots, T$ be the sampled activation trace of channel c feeding a delta-encoded matrix–vector product. The send-on-delta encoder [16, 17] keeps an anchor $\hat{a}$ (the last transmitted value), fires when $|a_c(t) - \hat{a}| > \theta$, and updates the anchor to $a_c(t)$ (anchor-on-event, preventing drift). Every event costs the fan-out of the weight column in MACs plus the associated memory traffic (Section 4).

Rice's formula and its discrete-time form. For a stationary process with spectral moments $\lambda_0 = \operatorname{Var} a$ and $\lambda_2 = \operatorname{Var} \dot{a}$, the mean rate of up- and down-crossings of level u is, under Gaussianity [21],

$$\mu(u) \;=\; \frac{1}{\pi}\sqrt{\frac{\lambda_2}{\lambda_0}}\,\exp\Big(-\frac{(u-m)^2}{2\lambda_0}\Big). \tag{1}$$

Sampled traces are sequences, not continuous paths: for a stationary Gaussian *sequence* with lag-1 autocorrelation ρ_1 the exact per-step crossing probability has the closed form

$$P\big[(a_t - u)(a_{t+1} - u) < 0\big] \;=\; 4\,\mathrm{T}\Big(\tfrac{u-m}{\sqrt{\lambda_0}}, \sqrt{\tfrac{1-\rho_1}{1+\rho_1}}\Big), \tag{2}$$

with T Owen's T function [19]; at $u = m$ this is the classical $\arccos(\rho_1)/\pi$ [12], and $\rho_1 \to 0$ recovers the iid value $2\Phi(\tilde{u})(1-\Phi(\tilde{u}))$. Eq. (1) with finite-difference moments underpredicts Eq. (2) as correlation weakens—by up to 8.8% at $\rho_1 = 0.1$ (Section 5)—so we use (1) as the headline Gaussian predictor, as the delta literature would, and (2) as the refined form (both are reported).

Differentiable estimator. The program's smoothed Monte-Carlo estimator [8] evaluates the co-area integrand $\delta_\epsilon(a - u)\,|\dot{a}|$ at sample points. On a sampled trace, the piecewise-linear segment $a_t \to a_{t+1}$ admits an *exact* smoothed count,

$$\hat{c}(u) \;=\; \frac{1}{(T-1)\,\Delta t}\sum_{t=1}^{T-1}\Big|\,\Phi\big(\tfrac{a_{t+1}-u}{\epsilon}\big) - \Phi\big(\tfrac{a_t-u}{\epsilon}\big)\Big|, \tag{3}$$

which converges to the hard sign-change count as $\epsilon \to 0$, remains differentiable in the trace values—hence in the network parameters—and is robust to steps that jump many ϵ in one Δt, where the midpoint co-area rule undercounts (Fig. 1b,c).

From crossings to delta events. Summing crossing rates over the implicit ladder of transmission levels spaced θ apart bounds the send-on-delta event rate by total variation [16]:

$$R(\theta) \;\le\; \min\big(\mathbb{E}|\Delta a|/(\theta\,\Delta t),\; 1/\Delta t\big), \qquad \mathbb{E}|\dot{a}| = \sqrt{2\lambda_2/\pi}\ \text{(Gaussian)}. \tag{4}$$

The bound is loose by exactly the hysteresis of the anchor (re-crossing a level without traveling θ) and by multi-level jumps. We measure this gap once, on synthetic processes, as a correction $\gamma(\theta/\sigma_\delta)$ with σ_δ the one-step increment scale (Fig. 2); for $\theta \le \sigma_\delta/2$ it is process-independent to 0.5% across four process families. On real activation traces, whose increments are heavy-tailed, the synthetic γ transfers imperfectly (up to 54% error; Section 6); the empirical route—simulating the encoder on cached calibration traces, which costs no extra network forwards—is what allocation uses.

One-sided crossing budget. The training objective transplants the Howe [7] mechanism to its native domain: per layer ℓ with hidden traces h_ℓ,

$$\mathcal{L}^{(\ell)}_{\text{budget}} \;=\; \operatorname{mean}_j\; \frac{\operatorname{relu}\big(\hat{c}_\ell(u_j) - b_j\big)^2}{(b_j + \bar{b})^2}, \qquad b_j = \rho\, c^{\text{cal}}_\ell(u_j), \tag{5}$$

with levels u_j on a quantile ladder of the layer's calibration activation distribution (the dense-ladder rule of Howe [9]), budgets a ρ-scaled copy of the calibration profile, and $\hat{c}$ the segment estimator (3). Within budget the loss and its gradient are exactly zero (one-sidedness; unit-tested), so compliant dynamics are untouched. The sampling-scale failure mode of Howe [9] does not arise here: traces are densely sampled 1-D sequences.

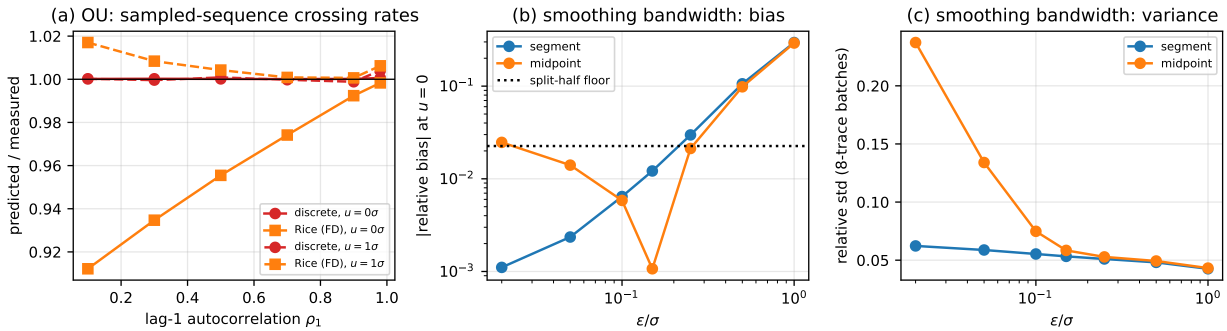

Figure 1: Estimator validation. (a) Sampled OU/AR(1) sequences: the discrete-time Gaussian form (2) is exact across correlation levels (max error 1.0%); the continuous Rice formula with finite-difference moments underpredicts by up to 8.8% as $\rho_1 \to 0$. (b,c) Smoothing bandwidth ϵ: the segment estimator (3) has <1% bias for $\epsilon \le 0.1\sigma$ at working trace lengths (256×100) with batch-to-batch variability already at the split-half noise floor (2.3%); the midpoint co-area rule trades a bias spike at small ϵ against inflated variance.

4 Experimental setup

Tasks and models. (i) **SC2**: Google Speech Commands v2 [23], 35-class keyword spotting, 40-bin log-mel features (25 ms window, 10 ms hop, T=101), 2-layer GRU [3] with 128 units, mean-pool, linear head; test accuracy 91.7%±0.11% over 8 seeds. (ii) **psMNIST**: row-sequential MNIST [13] (T=28 rows of 28 pixels), same GRU; 98.7%±0.05% over 3 seeds. (iii) **enwik8** (secondary): byte-level char-LM [15] on a 20M-character slice, 2-layer causal transformer (dim 128, 4 heads, FFN 512, T=256), 2.23 test bpc over 3 seeds. The delta-GRU follows Neil et al. [17] exactly (thresholds on input and hidden streams, anchor-on-event, incremental accumulators); at θ=0 it reproduces `torch.nn.GRU` to numerical precision (unit-tested). For the transformer, the input streams of the QKV, attention-output, and both FFN projections are delta-encoded along the sequence axis; attention score/value matmuls stay dense, as in the event-driven transformer literature. All experiments run on one RTX 3080 (10 GB).

Events and modeled energy. One event = one transmitted vector component = its weight-column fan-out in MACs plus the associated SRAM traffic. We report MAC-weighted event rates as fractions of the dense network, and modeled energy using Horowitz [6] 45 nm figures (FP32 MAC 4.6 pJ, 32-bit SRAM read 5 pJ, weights SRAM-resident). *The paper reports event counts and modeled energy, never measured watts.*

Discipline. Every number in this paper is machine-generated from committed result JSONs (`gen_paper_numbers.py`, `make_figures.py`); the regenerate-and-diff check passes at submission. All estimator and delta-cell properties are unit-tested (`pytest`, 14 tests). Gates were pre-registered in the project brief: GATE V (estimator vs closed forms, ≤3%), GATE P (calibration-split predictor within ∼10% on real networks), and the E4 kill condition (budget ≈ L1-delta at matched events must be reported as such).

5 E0: estimator validation (GATE V)

Random-phase-sinusoid Gaussian processes with *known* spectral moments: Rice (1) vs dense-grid hard counts at levels 0–1.5σ, 3 seeds, max relative error 1.9%. Sampled AR(1) (exact OU discretization, deliberately *not* mean-square differentiable): the discrete form (2) is exact to 1.0% while continuous Rice-from-FD-moments underpredicts by up to 8.8% at weak correlation—the bias regime real activation traces occupy (Fig. 1a). Deterministic multisines: segment estimator vs exact counts, 0.18%. **GATE V passes** (all ≤3%). Fig. 1b,c documents the ϵ trade-off once: we use $\epsilon = 0.15\sigma$ for training (bias <2%, variance at the noise floor) and hard counts for all measurement. Fig. 2 shows the hysteresis correction γ and its small-x universality.

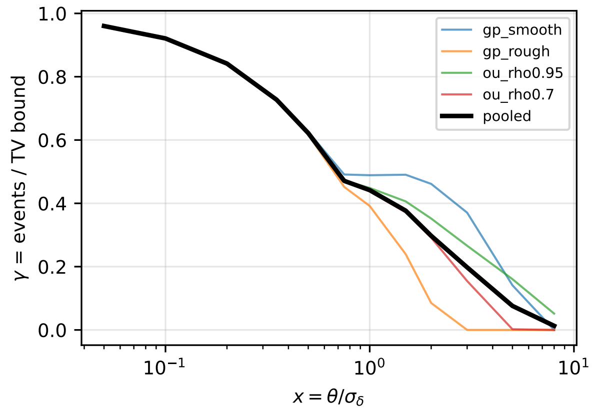

Figure 2: Send-on-delta correction. Measured event rate over the TV/θ bound, γ, vs $x = \theta/\sigma_\delta$ for four synthetic process families (3 seeds each). For $x \leq 0.5$ the curves collapse (spread 0.5%); beyond $x \approx 1$ the tail of the increment distribution takes over and process-dependence sets in.

6 E1: prediction on real networks (GATE P)

For each task and seed we record activation traces on a calibration split (2048 sequences) and predict per-channel crossing rates on a disjoint evaluation split at channel-standardized levels. Three predictors: (a) Rice (1) from fitted moments, (b) the empirical calibration profile, (c) Gaussian-iid (temporal structure discarded). Table 1 and Fig. 3:

(b) carries the paper: median error 3.1%/1.8%/2.9% on SC2/psMNIST/enwik8, worst stream 8.1%. **GATE P passes** on all three tasks. **(a) fails exactly where it should**: it holds to 8% on LayerNorm-fed transformer streams (excess kurtosis ≈0) and degrades to 75% on deep GRU states and 48% on GELU outputs (kurtosis up to several hundred); the failure is two-sided—sub-Gaussian channels (saturated, bimodal tanh states, negative excess kurtosis) break it as surely as heavy tails—and per-channel Rice error correlates with |excess kurtosis| at Spearman $\rho = 0.40/0.25/0.80$ (Fig. 8). This quantifies, rather than hand-waves, where Gaussian theory is usable. **(c) is off by 5.6×–17.3×** on GRU streams: temporal structure is the entire game.

Event rates. Deployed (closed-loop) event rates are predicted from base-network statistics by open-loop send-on-delta simulation on the cached calibration traces: max error 3.0% (SC2) and 4.2% (psMNIST) across all streams and thresholds in the regime where accuracy stays within a point of dense; prediction degrades only where the delta approximation itself breaks the network (psMNIST $x \geq 1$, accuracy <52%), i.e. outside any useful operating point (Fig. 4). The fully analytic route (Gaussian TV + synthetic γ) misses by up to 54% on real traces—their increments are heavy-tailed, violating the small-x universality of Fig. 2—which is why allocation uses the empirical curves. On enwik8, thresholding costs quality quickly (2.23→2.34 bpc already at x=0.1, 87% events): character streams carry little temporal redundancy at the feature level, unlike audio or image rows; prediction still works (max 0.8% error at $x \leq 0.1$), but delta deployment is not attractive there, consistent with event-driven transformers targeting video/event streams rather than text.

7 E2: analytic threshold allocation

Deployment requires per-stream thresholds $(\theta_{\text{in}}, \theta_{h_1}, \theta_{h_2})$. The incumbent tunes them by search, each configuration costing a validation evaluation. We instead record traces in *one* calibration pass, build per-stream rate curves $R_s(\theta)$ by open-loop simulation on the cached traces (no further network forwards), and invert them for a target budget: a single relative threshold x^* (with $\theta_s = x^* \sigma_{\delta,s}$) solved by bisection on the predicted aggregate curve; a proportional-share variant; and a single absolute θ (all use the predicted curves; they differ in how thresholds distribute across streams).

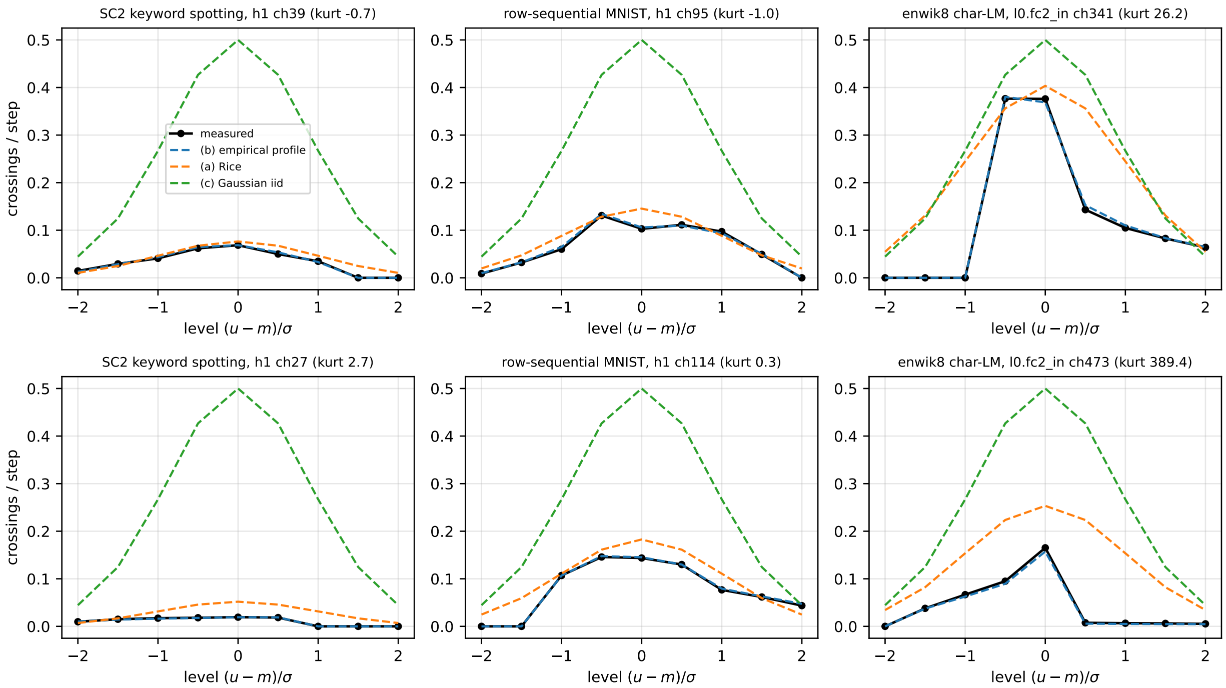

Figure 3: Per-channel crossing profiles (points: measured on held-out data; lines: predictors fitted on a disjoint calibration split). Top row: typical channels; bottom row: high-kurtosis channels. The empirical profile (b) tracks everything including the kurtosis-389 GELU channel (right); Rice (a) is good on near-Gaussian channels and systematically wrong on heavy-tailed ones; the iid baseline (c) is not in the running.

Fig. 5: on SC2 the realized event budgets land within 4% of target for budgets ≥20% of dense, and the analytic points sit on the 48-configuration random-search front. At the tightest budget that costs ≤1 accuracy point, the analytic allocation needs 29% of dense events vs 32% for the full 48-configuration search and 46% for a 6-configuration search—i.e. it matches the asymptotic sweep and dominates the cost-matched one at 48× lower tuning cost. On psMNIST the same holds at moderate budgets, while at aggressive budgets the joint search exploits per-stream asymmetries the analytic heuristics do not (42% vs 35%); the allocation-rule choice among our three variants is second-order everywhere—what replaces the sweep is the predicted curves themselves. In the accuracy-collapse regime, realized events undershoot targets (feedback reduces activity; consistent with Fig. 4); allocation there is meaningless anyway. As a modeled-energy anchor: at the 20%-of-dense operating point, SC2 keyword spotting runs at 90.5% accuracy for 30.3 μJ per decision vs 158.2 μJ dense (5.2× modeled reduction; Horowitz-table energy, not measured watts).

8 E3–E4: budget training vs null controls (negative)

Per task and seed, the base model is fine-tuned (5 epochs) with $\mathcal{L}_{\text{task}}+\lambda\,\mathcal{L}_{\text{budget}}$ (Eq. 5; $\rho \in \{0.7, 0.5, 0.35, 0.2\}$ the Pareto knob), or with the null controls: L1 on one-step deltas $|h_t - h_{t-1}|$, a spiking-style activity penalty $\overline{h^2}$, a *plain* fine-tune (no regularizer; isolates the effect of extra epochs), and the post-hoc sweep on the unmodified network. Every model is then evaluated closed-loop over a threshold sweep $\theta_s = x\,\sigma_{\delta,s}$ using *its own* calibration statistics (thresholds re-tuned per model, as deployment would), giving each model its own accuracy–events front.

The budget does what it promises—and it does not help. Compliance is real: the budget loss falls to near zero, crossing profiles drop as demanded, and hidden increment scales shrink 3.9× at ρ=0.2 (psMNIST). But at matched events the front never improves (Fig. 6): at the reference budget of 40% of dense events on psMNIST, post-hoc reaches 98.6% vs the budget family's 98.2% (paired difference -0.4%); on SC2 at 20%, 90.5% vs 90.4% (paired difference -0.1%, n=8 seeds, Wilcoxon p=0.312). The controls tie

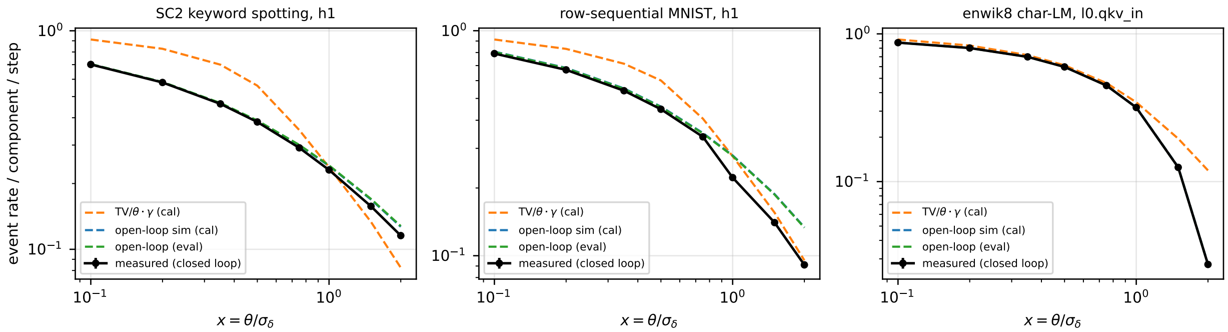

Figure 4: Delta-event rates, measured closed-loop (all streams thresholded at $\theta = x\sigma_\delta$) vs predictions from base-network statistics. Open-loop simulation on calibration traces (blue) tracks the deployed rates until the delta approximation itself degrades the network (psMNIST beyond $x \approx 0.75$); the analytic TV$/\theta \cdot \gamma$ curve (orange) inherits the synthetic γ's error on heavy-tailed real increments.

Table 1: Crossing-profile prediction on held-out data: median relative error and fraction of (channel, level) cells within 10%, over seeds; levels at $m_c + k\sigma_c$, $k \in [-2, 2]$, cells above the 0.005/step noise floor. "Floor" = split-half measurement floor.

task	stream	(b) empirical err	(b) empirical ≤10%	(a) Rice err	(a) Rice ≤10%	(c) iid err	(c) iid ≤10%	floor
SC2	input	7.5%	62%	14.0%	35%	561.9%	0%	2.9%
SC2	h1	3.1%	94%	36.7%	14%	927.7%	0%	2.4%
SC2	h2	2.3%	98%	73.1%	4%	1706.3%	0%	2.3%
row-sequential	input	2.7%	92%	59.4%	6%	233.7%	5%	1.5%
row-sequential	h1	1.8%	95%	24.4%	21%	259.3%	0%	1.7%
row-sequential	h2	1.6%	99%	33.6%	18%	498.2%	0%	1.9%
enwik8	l0.qkv_in	1.8%	92%	14.3%	36%	13.6%	41%	3.3%
enwik8	l0.out_in	7.7%	61%	37.0%	15%	359.4%	0%	11.6%
enwik8	l0.fc1_in	2.5%	89%	11.5%	45%	24.3%	23%	4.2%
enwik8	l0.fc2_in	3.5%	85%	34.2%	17%	52.9%	11%	5.3%
enwik8	l1.qkv_in	2.6%	90%	11.8%	44%	25.7%	20%	4.2%
enwik8	l1.out_in	3.8%	84%	14.8%	35%	96.4%	3%	6.2%
enwik8	l1.fc1_in	2.8%	90%	7.9%	60%	21.9%	20%	4.8%
enwik8	l1.fc2_in	3.0%	94%	45.5%	12%	60.8%	9%	5.1%

the post-hoc front (plain 98.7%, L1 98.7%, rate 98.6% on psMNIST)—so the pre-registered kill condition fires in its strongest form: not only does the crossing budget fail to beat L1-delta, *no* temporal-smoothness fine-tune beats doing nothing.

Mechanism: approximate scale-freeness. A send-on-delta encoder driven at $\theta = x\,\sigma_\delta$ has event rate $\approx \gamma(x)\,\mathrm{TV}/(x\,\sigma_\delta)$ with TV $\propto \sigma_\delta$: the scale cancels. Training that shrinks σ_δ therefore moves a model *along* its own front, not above it; only restructuring the increment distribution's *shape* could help, and Fig. 7 shows the budget compresses scale rather than shape. The flip side is a warning about evaluation methodology: at matched *absolute* θ the budget-trained model fires 36% fewer events (at 2.5% accuracy cost)—a fixed-threshold comparison, the natural default, would report this as a large win. On the matched-events front the same model is dominated by post-hoc thresholding of its own base. Fixed-θ comparisons overstate what smoothness training buys.

Reconciliation with the training-for-sparsity line. This is not a contradiction of prior training results so much as a reinterpretation of how they are read. The Delta Activation Layer of Yousefzadeh and Sifalakis [25] trains with an activation-change objective and reports an almost 3× temporal-sparsity improvement; such gains are measured at a *fixed* inference threshold (or fixed sparsification heuristic), which is precisely

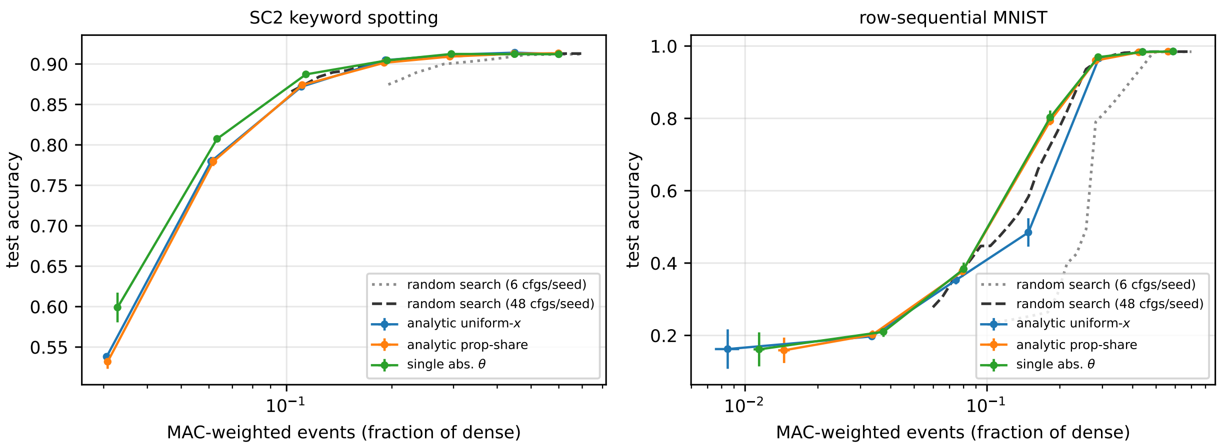

Figure 5: Analytic allocation vs threshold search (test accuracy vs MAC-weighted events, mean±sd over 3 seeds). The analytic operating points (one calibration pass, zero search) lie on the random-search front in the regime where accuracy is preserved; random search needs ∼48 configurations (48× the evaluation cost) to match them.

the matched-*absolute*-θ comparison that our scale-freeness analysis shows credits smoothness training with the 36% apparent saving measured above—a saving that vanishes once the threshold is re-tuned per model (the matched-*events* comparison, which is what deployment does). The actionable warning to our own subfield is therefore narrow and concrete: report event savings from a temporal-sparsity objective against a post-hoc baseline whose threshold is re-tuned to the same accuracy, not against the same model at a frozen threshold. Independently, the most recent temporal-sparsity work on large language models is training-*free*, thresholding a converged model post hoc [20]; the field's practice is already converging on the baseline our results endorse.

Overhead. The segment estimator's forward+backward adds a *fixed* 23 ms per layer per step (batch 256), independent of model width—it is an $O(NL)$ pass over trace points, not over parameters. On our deliberately tiny 128-unit GRU (task step 4.2 ms) this is a large relative cost (12× per step), but it is a fine-tuning-only expense that amortizes on realistically sized models, and Monte-Carlo point subsampling cuts it near-linearly (to 6 ms at 25% of points; the estimator *is* a Monte-Carlo integral, and Fig. 1c shows its variance is already at the noise floor). The negative result is not a cost story.

9 Discussion

What the theory buys. The validated chain is: activation statistics → crossing profiles (Rice where Gaussian, empirical profile in general) → event rates (open-loop simulation; TV bound with a universal small-x hysteresis correction on Gaussian-increment processes) → analytic threshold allocation that replaces sweeps. This is the piece the delta-network literature lacked: DeltaKWS-style deployments tune thresholds by silicon- or simulator-in-the-loop sweeps; a calibration pass plus curve inversion gets there for a few percent error at 48× less tuning compute on SC2.

What it does not buy. Training against the rate model (or any smoothness proxy) is Pareto-neutral at best under scale-adapted thresholds. This is consistent with, and explains, the delta literature's practice of training normally and thresholding afterwards: our results say that practice is not lazy, it is *correct* for these architectures and tasks. The scale-freeness argument suggests where a positive result might live: objectives that restructure increment *shape* (e.g., pushing traces toward piecewise-constant, genuinely sparse-in-time increments) rather than shrinking increment scale—EGRU-style hard gating [22] is one such mechanism, and bridging it to analytic rate models is open.

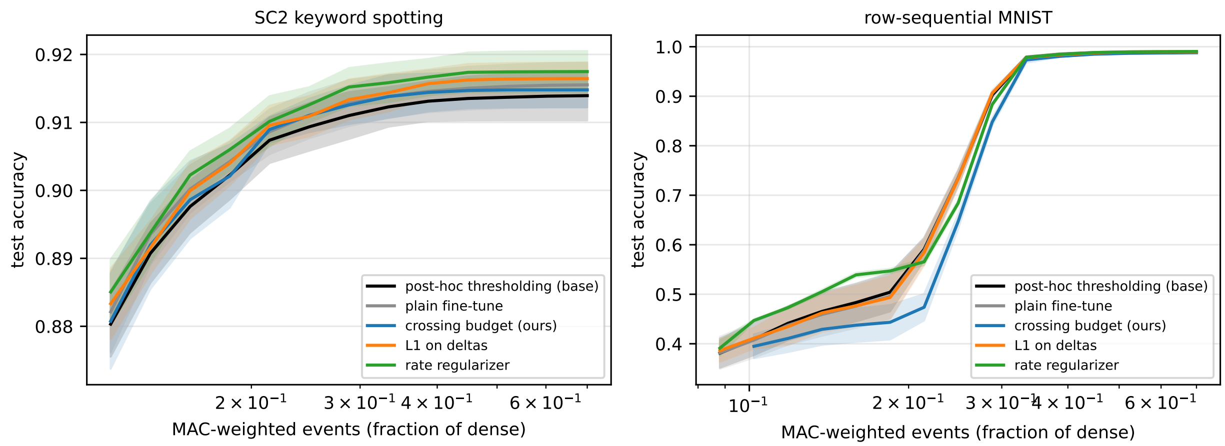

Figure 6: Training for temporal sparsity does not beat post-hoc thresholding. Accuracy–events Pareto envelopes (per family over its hyperparameter grid; mean±sd over 8 seeds on SC2, 3 on psMNIST). All fine-tuned families—crossing budget, L1-on-deltas, rate regularizer, plain fine-tune—sit on or below the post-hoc front of the unmodified base network.

Information per event (framing). One can read the event stream through Palm calculus: conditioning on event times, the relevant quantity is information per event (bits per joule under the energy model). Our negative says fine-tuning did not increase bits-per-event at matched accuracy; we flag this as framing, not a load-bearing claim.

Neuromorphic outlook. The prediction/allocation machinery is exactly what event-driven hardware needs at design time: given trace statistics from a workload corpus, event rates—hence energy and bandwidth envelopes—follow analytically, before silicon or even simulation. The modeled 5.2× energy reduction at 90.5% accuracy for always-on keyword spotting is the standard delta-network story [2], now with predicted rather than swept thresholds. We claim no measured silicon numbers.

10 What we do not claim

We do not claim spiking or spike-coding results (Spikformer/SpikeGPT are a different mechanism); we do not claim measured hardware energy (all energy is Horowitz-table modeled, weights SRAM-resident); we do not claim the training negative generalizes past the tested scale (GRUs to 128 units, transformers to 2 layers, three tasks) or to training-from-scratch with event costs in the loss from step zero; we do not claim the analytic γ transfers to heavy-tailed increments (it does not; measured at up to 54% error); and the Palm-calculus paragraph is framing only.

11 Limitations

Fine-tuning (5 epochs) rather than from-scratch training; one architecture family per task; the budget acts on hidden-state streams via pooled per-layer quantile ladders (per-channel budgets untested); the allocation heuristics are one-dimensional inversions, not joint optimizers, and lose to joint search at accuracy-destroying budgets on psMNIST; enwik8 quality under thresholding is evaluated on fixed windows (no KV-cache-aware decoding); event accounting charges each matvec's input stream independently (a shared encoder would shave the anchor-update term); and all energy numbers are model-based.

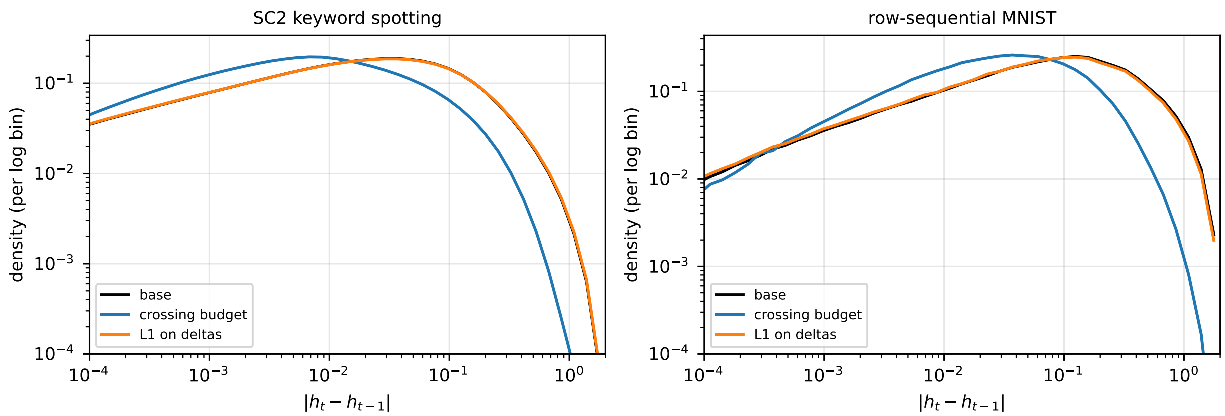

Figure 7: Mechanism check: increment histograms (density per log bin of $|h_t - h_{t-1}|$, mean over seeds). The crossing budget (blue) shifts the *entire* distribution toward zero—uniform scale compression, ≈one decade—while L1-on-deltas (orange) barely departs from the base network (black) at comparable fine-tuning strength. Neither changes the distribution's *shape*, which is what threshold re-scaling would need to find purchase on.

12 Reproducibility

Code, tests, experiment scripts, and every result JSON regenerate the paper: https://github.com/gunnerhowe/Research (directory `energy-effic`); archived at DOI: 10.5281/zenodo.21205335. Environment: Python 3.13.6, PyTorch 2.7.1+cu118, single RTX 3080 (10 GB), Windows 11; seeds 0–7 (SC2) / 0–2 (psMNIST, enwik8); `python -m pytest tests/ -q` (14 tests) and `python experiments/exp{0..4}*.py` reproduce `results/*.json`; `make_figures.py` and `gen_paper_numbers.py` regenerate every figure, table, and prose number; the regenerate-and-diff check passes at submission. Speech Commands v2 and MNIST download automatically; enwik8 from the Hutter corpus.

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A Discrete-time Gaussian crossing rate

For a stationary Gaussian sequence with mean m, variance λ_0, lag-1 correlation ρ_1, the pair (a_t, a_{t+1}) is bivariate normal; with $h = (u-m)/\sqrt{\lambda_0}$, $P[a_t \le u, a_{t+1} \le u] = \Phi(h) - 2\mathrm{T}(h,a)$ with $a = \sqrt{(1-\rho_1)/(1+\rho_1)}$ [19], giving the crossing probability $2[\Phi(h) - \Phi_2(h,h;\rho_1)] = 4\mathrm{T}(h,a)$ (Eq. 2). Sanity limits: $\rho_1 \to 0$ gives $2\Phi(h)(1-\Phi(h))$ (iid); $u = m$ gives $\arccos(\rho_1)/\pi$; both are unit-tested against AR(1) simulation (max error 1.0%). On real traces Eq. (2) tracks Eq. (1) closely (both fail together under non-Gaussianity): median errors differ by <2% on all streams, so the headline text reports (1).

B Kurtosis and the failure of the Gaussian predictor

Fig. 8 plots per-channel Rice error against excess kurtosis pooled over all streams, seeds, tasks: the V-shape shows the Gaussian predictor failing on *both* sides of Gaussianity.

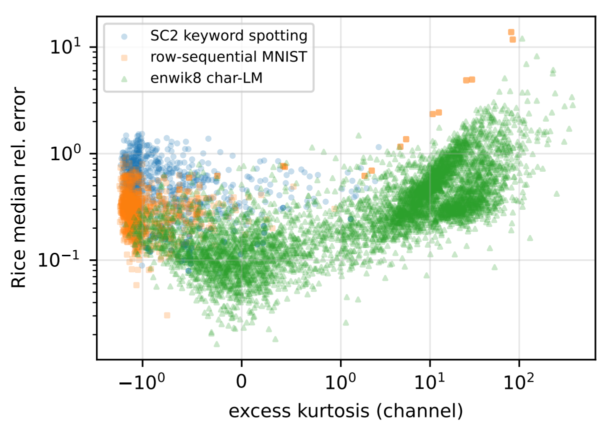

Figure 8: Per-channel Rice median relative error vs excess kurtosis (symlog axis). The error is minimal near zero kurtosis and rises two-sidedly; Spearman ρ of error vs |kurtosis|: 0.40 (SC2), 0.25 (psMNIST), 0.80 (enwik8).

C enwik8 under thresholding

Test bpc at uniform relative threshold x: 2.23 (dense) $\to$ 2.34 (x=0.1) $\to$ 3.69 (x=0.35). The delta-transformer's useful regime is $x \lesssim 0.1$; within it, event prediction errors are $\le$0.8%.

D Reproduction commands

```
python -m pytest tests/ -q
python experiments/exp0_validation.py
python experiments/train_base.py --task all
python experiments/train_base.py --task sc2 --seeds 3,4,5,6,7
python experiments/exp1_prediction.py --task all
```

```
python experiments/exp2_allocation.py
python experiments/exp3_budget_training.py
python experiments/exp3b_absolute_theta.py
python experiments/exp4_controls.py
python experiments/bench_timing.py
python experiments/make_figures.py
python paper/gen_paper_numbers.py
```